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AI-Powered IoT-Based Microclimate Control System for Smart Indoor Farming

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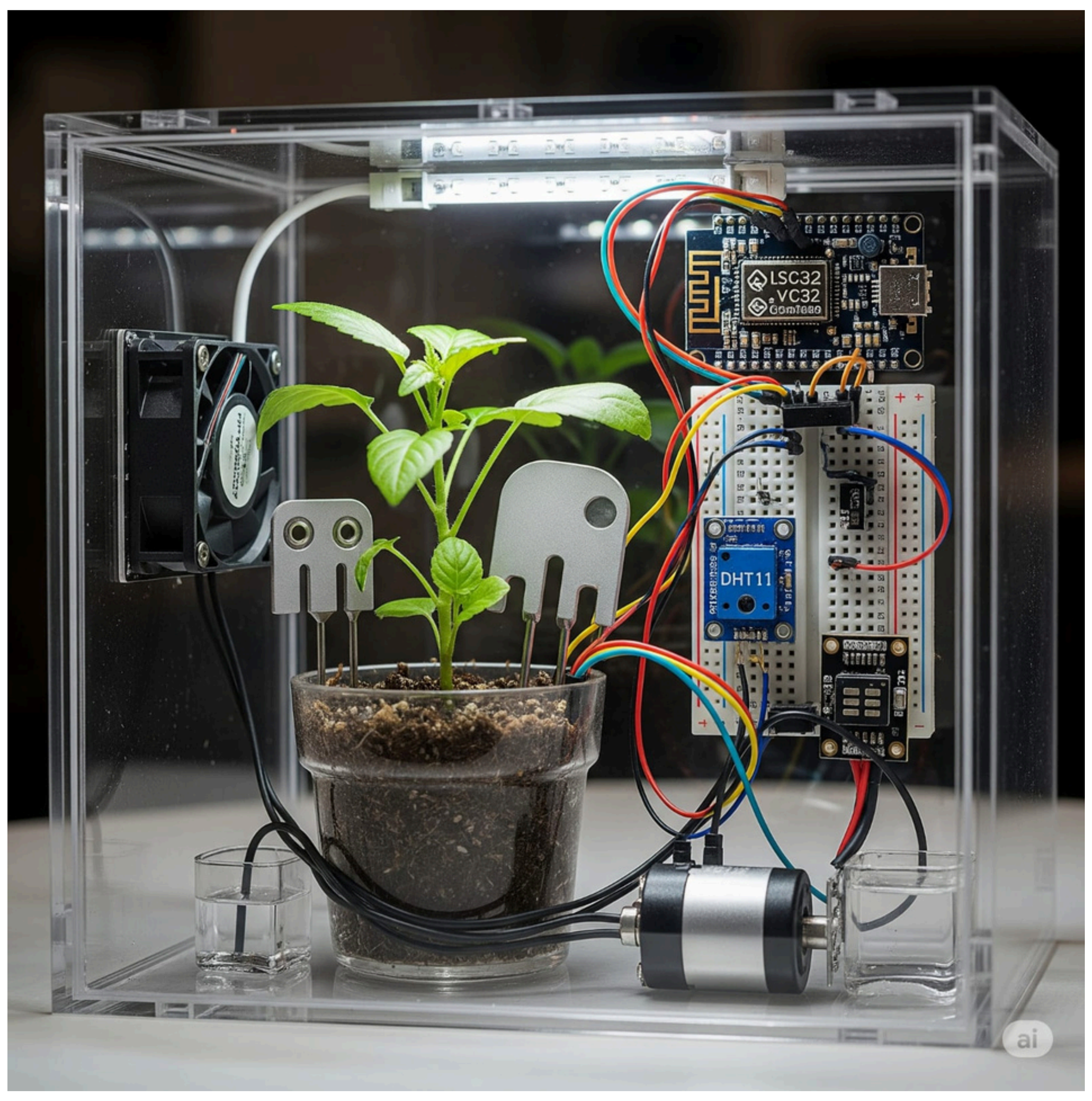
Introduction

Food security a critical global challenge. While indoor farming offers a solution, manually managing microclimatic parameters like temperature, humidity, and soil moisture is labor-intensive and inconsistent. This project leverages the advancements in Artificial Intelligence (AI) and the Internet of Things (IoT) to create a solution that continuously captures environmental data and makes real-time decisions to optimize crop growth.

Device Type	Component	Function
Sensors	DHT11/DHT22	Temperature & Humidity sensing
	LDR/Light Sensor	Light intensity measurement
	Soil Moisture Sensor	Root-zone water availability
Actuators	Ventilation Fan	Controls air circulation and temperature
	Artificial Light	Provides supplemental lighting
	Water Pump	Supplies water when soil dries



Workflow Stage	Current Model: Threshold-Based	Proposed Model: AI-Integrated (Reinforcement Learning)
Sensing	IoT sensors (DHT11, LDR, Soil Moisture) continuously record real-time environmental data.	IoT sensors (DHT11, LDR, Soil Moisture) continuously record real-time environmental data.
Initial Learning	Not applicable. The system operates on static, pre-set thresholds.	Users manually operate actuators (humidifier, fan, light, pump) for a few days, and sensors record these decisions alongside environmental conditions.
Decision & Adaptation	The system compares sensor data to pre-set thresholds and triggers actuators when a threshold is crossed.	An RL agent identifies patterns in user interventions and learns optimal control strategies. The trained system then autonomously regulates actuators in real-time to optimize growth.
User Trust & Transparency	Not applicable. Actions are based on simple rules.	Every action taken by the AI is accompanied by an explanation (e.g., "Fan switched ON because humidity exceeded safe range for crop growth") to ensure user trust.

Results (Preliminary Stage)

Sensor-Actuator Response: The threshold-based box setup successfully demonstrated environmental monitoring and actuation. Environmental parameters inside the box stabilized within the desired range, confirming feasibility.

Plant Health: The selected plant maintained steady growth conditions indoors, validating that the system can replicate favorable conditions.

User Interaction: Early tests indicate that the system can react quickly to environmental deviations, reducing manual intervention.

Conclusion

The project proves a feasible, smart indoor farming system using IoT and AI. The current stable prototype can evolve with RL for adaptive learning and XAI for user trust, promoting sustainable, scalable agriculture.

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